Quantum Field Theory II     Week 3     Homework

$\boxed{\text{Exercise 3.6}}$

$k, \bar{k}, \varepsilon^{(1)}, \varepsilon^{(2)}$ form a basis and we can also express

$P_{\mu\nu}$ in terms of $g_{\mu\nu}$, $k_\mu k_\nu$, $\bar{k}_\mu \bar{k}_\nu$, $\bar{k}_\mu k_\nu$, $k_\mu \bar{k}_\nu$

$$P_{\mu\nu} = A g_{\mu\nu} + B \frac{k_\mu k_\nu}{k\cdot\bar{k}} + C \frac{\bar{k}_\mu \bar{k}_\nu}{k\cdot\bar{k}} + D \frac{\bar{k}_\mu k_\nu}{k\cdot\bar{k}} + E \frac{k_\mu \bar{k}_\nu}{k\cdot\bar{k}}$$

Here we have normalized such that $A, B, C, D, E$ are dimensionless

Because $k, \bar{k}, \varepsilon^{(1)}, \varepsilon^{(2)}$ form a basis they are orthogonal i.e. $\varepsilon^{(1)}_\mu k^\mu = 0$ also $k^2 = \bar{k}^2 = 0$

Let's find out the relationships between the constants $A, B, C, D$ using projection of $P_{\mu\nu}$ along the basis vectors:

$\boxed{k_\mu g^{\mu\nu} = k^\nu}$

We start with projection along $g_{\mu\nu}$:

$$g_{\mu\nu} P^{\mu\nu} = 4A + D \underbrace{g_{\mu\nu} \frac{\bar{k}_\mu k_\nu}{k\cdot\bar{k}}}_{=1} + E \underbrace{g_{\mu\nu} \frac{k_\mu \bar{k}_\nu}{k\cdot\bar{k}}}_{=1} = 4A + D + E = g_{\mu\nu} P^{\mu\nu}$$

$$k_\mu P^\mu = A k_\mu g^{\mu\nu} + 0 + C \frac{k\cdot\bar{k}}{k\cdot\bar{k}} \bar{k}^\nu + D k^\nu + 0 = A k^\nu + C \bar{k}^\nu + D k^\nu$$

$$\bar{k}_\mu P^{\mu\nu} = A \bar{k}_\mu g^{\mu\nu} + B k^\nu + 0 + 0 + E \bar{k}^\nu = A \bar{k}^\nu + B k^\nu + E \bar{k}^\nu$$

$$k_\nu P^{\mu\nu} = A k_\nu g^{\mu\nu} + 0 + C \bar{k}^\mu + 0 + E k^\mu = A k^\mu + C \bar{k}^\mu + E k^\mu$$

$$\bar{k}_\nu P^{\mu\nu} = A \bar{k}_\nu g^{\mu\nu} + B k^\mu + 0 + D \bar{k}^\mu + 0 = A \bar{k}^\mu + B k^\mu + D \bar{k}^\mu$$

$$k_\nu \bar{k}_\mu P^{\mu\nu} = A k_\nu \bar{k}_\mu g^{\mu\nu} + E\, k\cdot\bar{k} = A\, k\cdot\bar{k} + E\, k\cdot\bar{k}$$

$$k_\mu \bar{k}_\nu P^{\mu\nu} = A k_\mu \bar{k}_\nu g^{\mu\nu} + D k_\mu \bar{k}^\mu = (A + D)\, k\cdot\bar{k}$$

Should be same under change of indices $\Rightarrow E = D$

$$k_\mu k_\nu P^{\mu\nu} = C\, k\cdot\bar{k} \quad \text{and} \quad \bar{k}_\mu \bar{k}_\nu P^{\mu\nu} = B\, k\cdot\bar{k} = (k^\rho k^\sigma g_{\rho\mu} g_{\rho\sigma} P^{\mu\nu})$$

$$= (k^\rho k^\sigma P_{\rho\sigma}) = (k_\rho k_\sigma P^{\rho\sigma}) = C \cdot k\cdot\bar{k}$$

$\Rightarrow B = C$

Additionally $k_\mu P^{\mu\nu} = k_\mu \sum_\lambda \varepsilon^\mu_\lambda \varepsilon^{\nu *}_\lambda = 0$ // since $k_\mu \varepsilon^\mu_\lambda = 0$

$\Rightarrow \bar{k}_\mu P^{\mu\nu} = 0$

Now we can relate the results of projeting $P^{\mu\nu}$ in 2-different way:

From $g_{\mu\nu} P^{\mu\nu} = 4A + D + E = 4A + 2D$ and $g_{\mu\nu} P^{\mu\nu} = g_{\mu\nu} \sum_\lambda \varepsilon^\mu_\lambda \varepsilon^{\nu *}_\lambda = \sum_\lambda \underbrace{\varepsilon \cdot \varepsilon}_{=-1} = -2$

we get $4A + 2D = -2 \Rightarrow 2A + D = -1$

From $k_\mu P^{\mu\nu} = 0 = (A+D) k^\nu + C \bar{k}^\nu$ we get $A = -D$ and $C = B = 0$

$$\Rightarrow \quad 2A + D = -1 \Leftrightarrow 2A - A = -1 \Leftrightarrow A = -1 \Rightarrow D = 1$$

Now we apply the results to the general $P$:

$$P_{\mu\nu} = \sum_\lambda \varepsilon^\lambda_\mu \varepsilon^{\lambda *}_\nu = A g_{\mu\nu} + B \left[ \frac{k_\mu k_\nu + \bar{k}_\mu \bar{k}_\nu}{k \cdot \bar{k}} \right] + D \left[ \frac{\bar{k}_\mu k_\nu + k_\mu \bar{k}_\nu}{k \cdot \bar{k}} \right]$$

$$\Rightarrow \quad P_{\mu\nu} = \sum_\lambda \varepsilon^\lambda_\mu \varepsilon^{\lambda *}_\nu = -g_{\mu\nu} + \frac{\bar{k}_\mu k_\nu + k_\mu \bar{k}_\nu}{k \cdot \bar{k}} \qquad \square$$

**Exercise 3.7** Consider pure electrodynamics,

$$S[A] = \int d^4x \left( -\frac{1}{4} F_{\mu\nu} F^{\mu\nu} \right)$$

using the gauge fixing condition $G_w[A] = \partial_\mu A^\mu + \frac{1}{2} \eta A_\mu A^\mu - w$,

where $w(x) = w$ is an arbitrary function.

The FP-determinant is $\Delta_{FP} = \det \left( \frac{\delta G_w[A^\alpha]}{\delta \alpha} \right)$

$$\Rightarrow \quad 1 = \int D\alpha(x)\, \delta(G(A^\alpha)) \det \left( \frac{\delta G_w[A^\alpha]}{\delta \alpha} \right)$$

We insert this into the definition of path integral to get

$$\int D\alpha \int DA_\mu\, \Delta_{FP}(A_\mu) \delta(G[A^\alpha_\mu])\, e^{iS[A]} = \int D\alpha \int DA_\mu\, e^{iS[A]} \delta(G[A^\alpha]) \det \left( \frac{\delta G[A^\alpha]}{\delta \alpha} \right)$$

$$= \int D\alpha \int DA_\mu\, S^{iS[A]}\, \delta\left( \partial_\mu A^{\alpha\mu} + \frac{1}{2} \eta A^\alpha_\mu A^{\alpha\mu} - w \right) \det \left( \frac{\delta G[A^\alpha]}{\delta \alpha} \right)$$

This integral is true for any $w(x)$ and $\det \left( \frac{\delta G[A^\alpha]}{\delta \alpha} \right)$ is independent of of $w$ so we can multiply by normalization constant and at the same time integrate one $w(x)$ with a Gaussian weight:

$$\Rightarrow N(\xi)\int D\omega \, \exp\left[-i\int d^4x \frac{\omega^2}{2\xi}\right]\int D\alpha \int DA_\mu \, e^{iS[A]} \, \delta(\partial_\mu A^{\alpha\mu} + \tfrac{1}{2}\eta A^\alpha_\mu A^{\alpha\mu} - \omega(x)) \det\left(\frac{\delta G[A^\alpha]}{\delta\alpha}\right)$$

$$= N(\xi)\int D\alpha \int DA_\mu \, e^{iS[A]} \exp\left[-i\int d^4x \frac{1}{2\xi}\left[\partial_\mu A^{\alpha\mu} + \tfrac{1}{2}\eta A^\alpha_\mu A^{\alpha\mu}\right]^2\right] \det\left[\frac{\delta G[A^\alpha]}{\delta\alpha}\right]$$

Since the Lagrangian is invariant under gauge transformation we can replace $A$ in $e^{iS[A]}$ by $A^\alpha$. This doesn't chang the measure since $DA_\mu = DA^\alpha_\mu$ since $A \to A^\alpha$ is a simple shift.

$$\Rightarrow \; = N(\xi)\int D\alpha \int DA_\mu \, e^{iS[A]} \exp\left[-i\int d^4x \frac{1}{2\xi}\left[\partial_\mu A^\mu + \tfrac{1}{2}\eta A_\mu A^\mu\right]^2\right] \det\left[\frac{\delta G[A^\alpha]}{\delta\alpha}\right]$$

Now we use identity $\int D\bar{c}Dc \, \exp\left[-i\int d^4x \, \bar{c}\left(\frac{\delta G[A^\alpha]}{\delta\alpha}\right)c\right] = \det\left[\frac{\delta G[A^\alpha]}{\delta\alpha}\right]$

Which gives us:

$$N(\xi)\int D\alpha \, DA_\mu \, D\bar{c}\, Dc \, \exp\left[i\int d^4x \, \mathcal{L}_{QED} - \frac{1}{2\xi}(\partial_\mu A^\mu + \tfrac{1}{2}\eta A_\mu A^\mu)^2 - \bar{c}\left(\frac{\delta G[A^\alpha]}{\delta\alpha}\right)\bar{c}\right]$$

Let's now calculate $\frac{\delta G[A^\alpha]}{\delta\alpha}$ : with $G[A^\alpha] = \partial_\mu A^{\alpha\mu} + \frac{1}{2}\eta A^\alpha_\mu A^{\alpha\mu} - \omega$ $\;//\; A^\alpha_\mu = A_\mu + \frac{1}{e}\partial_\mu\alpha$

$$\frac{\delta G[A^\alpha]}{\delta\alpha} = \frac{\delta}{\delta\alpha}\left[\partial_\mu A^\mu + \tfrac{1}{e}\partial_\mu\partial^\mu\alpha + \tfrac{1}{2}\eta\left[(A_\mu + \tfrac{1}{e}\partial_\mu\alpha)(A^\mu + \tfrac{1}{e}\partial^\mu\alpha)\right]\right]$$

$$= \frac{1}{e}\partial^2 + \frac{1}{2}\eta\frac{\delta}{\delta\alpha}\Big[A_\mu A^\mu + \underbrace{\tfrac{1}{e}(\partial_\mu\alpha)A^\mu + \tfrac{1}{e}A_\mu(\partial^\mu\alpha)}_{=\frac{2}{e}A_\mu(\partial^\mu\alpha)} + \tfrac{1}{e^2}\underbrace{\partial_\mu\alpha\,\partial^\mu\alpha}_{\frac{1}{2}\partial^2\alpha^2}\Big]$$

$\partial_\mu\alpha\,\partial^\mu\alpha = \frac{1}{2}\partial_\mu\partial^\mu\alpha^2 = \frac{1}{2}\partial^2\alpha^2$

$$= \underline{\frac{1}{e}\partial^2 + \frac{1}{e}\eta A_\mu\partial^\mu + \frac{1}{2e^2}\eta\,\partial^2\alpha}$$

, we have $\alpha$-dependence because the Gauge is non-linear

Our Path integral has become:

Standard QED-Lagrangian

$$\Rightarrow F \equiv N(\xi)\int D\alpha \, DA_\mu \, D\bar{c}\, Dc \, \exp\Big[i\int d^4x \, \mathcal{L}_{QED} - \frac{1}{2\xi}(\partial_\mu A^\mu + \tfrac{1}{2}\eta A_\mu A^\mu)^2 - \bar{c}\left(\tfrac{1}{e}\partial^2\left[1 + \tfrac{1}{2e}\eta\alpha\right] + \tfrac{1}{e}\eta A_\mu\partial^\mu\right)\bar{c}\Big]$$

With the Help of term $\frac{1}{2\xi}(\partial_\mu A^\mu + \frac{1}{2}\eta A_\mu A^\mu)^2$ we can construct a photon Propagator

and $\frac{1}{e}\partial^2[1 + \frac{1}{2}\eta\alpha]$ is responsible for ghost propagator and $\frac{1}{e}\eta A_\mu\partial^\mu$ is responsible a interaction between photon field and Ghost-field. Let's now write just the Lagrangian part and read off the propagators :

$$\mathcal{L} = \mathcal{L}_{QED} - \frac{1}{2\xi}\left(\partial_\mu A^\mu + \frac{1}{2}\eta A_\mu A^\mu\right)^2 - \bar{c}\left[\partial^2\left(1+\frac{1}{2e}\eta\alpha\right) + \eta A_\mu \partial^\mu\right]c$$

Here we have absorbe one of the $\frac{1}{e}$ - term into $\bar{c}$ and $c$ definition.

$$\mathcal{L} = \mathcal{L}_D + \mathcal{L}_{int} + A_\mu\left(\partial^2 g^{\mu\nu} - \partial^\mu\partial^\nu\right)A_\nu - \frac{1}{2\xi}\left(\partial_\mu A^\mu + \frac{1}{2}\eta A_\mu A^\mu\right)^2 - \bar{c}\left[\partial^2\left(1+\frac{1}{2e}\eta\alpha\right) + \eta A_\mu\partial^\mu\right]c$$

here $\mathcal{L}_D = \mathcal{L}_{Dirac}$ and $\mathcal{L}_{int}$ is interaction Lagrangian

The photon only part is :

$$A_\mu\left(\partial^2 g^{\mu\nu} - \partial^\mu\partial^\nu\right)A_\nu - \frac{1}{2\xi}\left(\underbrace{\partial_\mu A^\mu}_{=A_\mu\partial^\mu} + \frac{1}{2}\eta A_\mu A^\mu\right)\left(\partial^\nu A_\nu + \frac{1}{2}\eta \underbrace{A_\nu A^\nu}_{=A^\nu A_\nu}\right)$$

let's absorb 2 into definition of $\xi$ $\Rightarrow$ $2\xi \to \xi$

$$= A_\mu\left(\partial^2 g^{\mu\nu} - \partial^\mu\partial^\nu\right)A_\nu - \frac{1}{2\xi}\left(A_\mu\left[\partial^\mu\partial^\nu + \frac{\eta}{2}\partial^\mu A^\nu\right]A_\nu + \frac{1}{2}\eta A_\mu\left[A^\mu\partial^\nu + \frac{1}{2}\eta A^\mu A^\nu\right]A_\nu\right)$$

$$= A_\mu\left[\partial^2 g^{\mu\nu} - \partial^\mu\partial^\nu - \frac{1}{\xi}\partial^\mu\partial^\nu - \frac{1}{2\xi}\eta\,\partial^\mu A^\nu - \frac{1}{2\xi}\eta A^\mu\partial^\nu - \frac{1}{4\xi}\eta^2 A^\mu A^\nu\right]A_\nu$$

$$= A_\mu\left[\partial^2 g^{\mu\nu} - \left(1-\frac{1}{\xi}\right)\partial^\mu\partial^\nu\right]A_\nu - \frac{\eta}{2\xi}A_\mu\left[\partial^\mu A^\nu - A^\mu\partial^\nu\right]A_\nu - \frac{1}{4\xi}\eta^2 \underbrace{A_\mu A^\mu A^\nu A_\nu}_{=A^4}$$

$$= \underbrace{A_\mu\left[\partial^2 g^{\mu\nu} - \left(1-\frac{1}{\xi}\right)\partial^\mu\partial^\nu\right]A_\nu}_{\text{This term lead to the propagator and this is exactly same as when the gauge-fixing was linear in } A_\mu} \underbrace{- \frac{\eta}{2\xi}A_\mu\left[\partial^\mu A^\nu - A^\mu\partial^\nu\right]A_\nu - \frac{1}{4\xi}\eta^2 A^4}_{\text{Self-interaction terms that arise from non-linear gauge-fixing}}$$

$\Rightarrow$ The photon propagator is still the same:

$$\tilde{D}_F(k) = \frac{-i}{k^2+i\varepsilon}\left(g^{\mu\nu} - (1-\xi)\frac{k^\mu k^\nu}{k^2}\right)$$

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The ghost Lagrangian : $\mathcal{L}_{ghost} = \bar{c}\left[-\partial^2\left[1+\frac{1}{2e}\eta\alpha\right] - \eta A_\mu\partial^\mu\right]c$

We can read from this lagrangian that the ghost propagator is $i\,\dfrac{1+\frac{1}{2e}\eta\alpha}{p^2}$

$\Rightarrow$ - - - - < - - - - $:= i\,\dfrac{1+\frac{\eta}{2e}\alpha}{p^2}$ , so the ghost propagator depends on $\alpha$ that is a gauge - transformation variable.

$$\boxed{\begin{array}{l}\text{The full QED Lagrangian in this gauge-fixing is}\\ \mathcal{L} = \bar{\Psi}(i\not{D}-m)\Psi - \frac{1}{4}F_{\mu\nu}F^{\mu\nu} - \frac{1}{2\xi}\left(\partial_\mu A^\mu + \frac{1}{2}\eta A_\mu A^\mu\right)^2 - \bar{c}\left[\partial^2\left(1+\frac{\eta\alpha}{2e}\right) + \eta A_\mu\partial^\mu\right]c\end{array}}$$

Exercise 3.8

$$\mathcal{L} = \mathcal{L}_{QED} - \frac{1}{2\xi}(\partial_\mu A^\mu)^2 + \partial^\mu \bar{c}\, \partial_\mu c$$

where $\mathcal{L}_{QED}$ is $\mathcal{L}_{QED} = \bar{\Psi}(i\not{D} - m)\Psi - \frac{1}{4}F_{\mu\nu}F^{\mu\nu}$ , with $D_\mu = (\partial_\mu + ieA_\mu)$

$$\Rightarrow \mathcal{L} = \bar{\Psi}(i\not{\partial} - m)\Psi - \frac{1}{4}F_{\mu\nu}F^{\mu\nu} - e\bar{\Psi}\gamma^\mu\Psi A_\mu - \frac{1}{2\xi}(\partial_\mu A^\mu) + \partial^\mu \bar{c}\,\partial_\mu c$$

Let's start doing the transformation part by part: first we only transform $A_\mu$ then $c$ and $\bar{c}$ then $\Psi$ and $\bar{\Psi}$ and after this we have done the whole transformation:

The transformation we are aiming to do is ($\delta\eta$ is Grassmann parameter)

$$\begin{aligned} A_\mu(x) &\to A'_\mu(x) = A_\mu(x) + \partial_\mu c(x)\,\delta\eta \\ c(x) &\to c'(x) = c(x) \\ \bar{c}(x) &\to \bar{c}'(x) = \bar{c}(x) + \frac{1}{\xi}\partial^\mu A_\mu(x)\,\delta\eta \end{aligned} \qquad \begin{aligned} \Psi(x) &\to \Psi'(x) = \Psi(x) - iec(x)\,\delta\eta\,\Psi(x) \\ \bar{\Psi}(x) &\to \bar{\Psi}(x) = \bar{\Psi}(x) + ie\bar{\Psi}(x)c(x)\,\delta\eta \end{aligned}$$

Under this transformation the Lagrangian becomes:

$$\Rightarrow \mathcal{L} \to \mathcal{L}' = \bar{\Psi}'(i\not{\partial} - m)\Psi' - \frac{1}{4}F'_{\mu\nu}F'^{\mu\nu} - e\bar{\Psi}'\gamma^\mu\Psi' A'_\mu - \frac{1}{2\xi}(\partial_\mu A'^\mu)^2 + \partial^\mu\bar{c}\,\partial_\mu c$$

First let's only expand $A'_\mu$ in terms of $A_\mu$

$$\Rightarrow F'_{\mu\nu} = \partial_\mu A'_\nu - \partial_\nu A'_\mu = F_{\mu\nu} + \partial_\mu\partial_\nu c\,\delta\eta - \partial_\nu\partial_\mu c\,\delta\eta$$

$$\Rightarrow F'_{\mu\nu}F'^{\mu\nu} = F_{\mu\nu}F^{\mu\nu} + F_{\mu\nu}\partial^\mu\partial^\nu c\,\delta\eta - F_{\mu\nu}\partial^\nu\partial^\mu c\,\delta\eta + \partial_\mu\partial_\nu c\,\delta\eta\,F^{\mu\nu} - \partial_\nu\partial_\mu c\,\delta\eta\,F^{\mu\nu}$$

All the other terms are 0 since they are proportional to $(\delta\eta)^2 = 0$

We can chang the order of derivation freely because for a smooth-enough function $\partial_\nu\partial_\mu f(x) = \partial_\mu\partial_\nu f(x)$, and we assume all our function are smooth

$$\Rightarrow F_{\mu\nu}[\partial^\mu\partial^\nu c\,\delta\eta - \partial^\nu\partial^\mu c\,\delta\eta] = 0 \quad \text{and} \quad [\partial_\mu\partial_\nu - \partial_\nu\partial_\mu]c\,\delta\eta\,F^{\mu\nu} = 0$$

$$\Rightarrow F'_{\mu\nu}F'^{\mu\nu} = F_{\mu\nu}F^{\mu\nu} \quad \Rightarrow \quad F_{\mu\nu}F^{\mu\nu} \text{ is invariant under the transformation}$$

For term $e\bar{\Psi}'\gamma^\mu\Psi' A'_\mu$ we get $e\bar{\Psi}'\gamma^\mu\Psi' A'_\mu = e\bar{\Psi}'\gamma^\mu\Psi' A_\mu + e\bar{\Psi}\gamma^\mu\Psi'\partial_\mu c\,\delta\eta$

and for $(\partial_\mu A'^\mu)^2$ we get $(\partial_\mu A'^\mu)^2 = (\partial_\mu A^\mu + \partial_\mu\partial^\mu c\,\delta\eta)^2 = (\partial_\mu A^\mu)^2 + \partial_\mu A^\mu\partial^2 c\,\delta\eta + \partial^2 c\,\delta\eta\,\partial_\mu A^\mu$

$$\Rightarrow (\partial_\mu A'^\mu)^2 = (\partial_\mu A^\mu)^2 + \partial_\mu A^\mu\partial^2 c\,\delta\eta + \partial^2 c\,\delta\eta\,\partial_\mu A^\mu$$

So our Lagrangian is now: $\mathcal{L}' = \bar{\Psi}'(i\not{\partial} - m)\Psi' - \frac{1}{4}F_{\mu\nu}F^{\mu\nu} - e\bar{\Psi}'\gamma^\mu\Psi' A_\mu + e\bar{\Psi}\gamma^\mu\Psi'\partial_\mu c\,\delta\eta$
$- \frac{1}{2\xi}(\partial_\mu A^\mu)^2 - \frac{1}{2\xi}[(\partial_\mu A^\mu)(\partial^2 c\,\delta\eta) + (\partial^2 c\,\delta\eta)(\partial_\mu A^\mu)] + \partial^\mu\bar{c}\,\partial_\mu c$

Now let's expand $\bar{c}'$ and $c'$

let's chang the indices to not get confused

$$\Rightarrow \quad \partial^\mu \bar{c}' \partial_\mu c' = \partial^\mu \left(\bar{c} + \frac{1}{\xi}\partial^\mu A_\mu(x)\delta\eta\right)\partial_\mu c = \partial^\mu \bar{c}\,\partial_\mu c + \frac{1}{\xi}\underbrace{\partial^\mu(\partial^\nu A_\nu \delta\eta)\partial_\mu c}$$

$$= \partial^\mu\left[\partial^\nu A_\nu \delta\eta\, \partial_\mu c\right] - (\partial^\nu A_\nu)\delta\eta\, \partial^2 c$$

$$\Rightarrow \mathcal{L}' = \bar{\Psi}'(i\gamma - m)\Psi' - \frac{1}{4}F_{\mu\nu}F^{\mu\nu} - e\bar{\Psi}'\gamma^\mu\Psi' A_\mu - e\bar{\Psi}'\gamma^\mu\Psi'\,\partial_\mu c\,\delta\eta - \frac{1}{2\xi}(\partial_\mu A^\mu)^2$$

$$-\frac{1}{2\xi}\Big[\underbrace{(\partial_\mu A^\mu)(\partial^2 c\,\delta\eta) + (\partial^2 c\,\delta\eta)(\partial_\mu A^\mu)}_{=2(\partial_\nu A^\nu)\partial^2 c\,\delta\eta}\Big] + \partial^\mu \bar{c}\,\partial_\mu c + \frac{1}{\xi}\left(\partial^\mu[\partial^\nu A_\nu \delta\eta\,\partial_\mu c] - (\partial^\nu A_\nu)\delta\eta\,\partial^2 c\right)$$

The term $\partial^\mu[\partial^\nu A_\nu \delta\eta\,\partial_\mu c]$ is a boundary term so it vanishes in the path integral in the exponent under the integral $\int d^4x$, $\Rightarrow$ no actual contribution to the Lagrangian $\Rightarrow$ we can set it to zero.

$$\Rightarrow \mathcal{L}' = \bar{\Psi}'(i\gamma - m)\Psi' - \frac{1}{4}F_{\mu\nu}F^{\mu\nu} - e\bar{\Psi}'\gamma^\mu\Psi' A_\mu - e\bar{\Psi}'\gamma^\mu\Psi'\,\partial_\mu c\,\delta\eta - \frac{1}{2\xi}(\partial_\mu A^\mu)^2 + \partial^\mu \bar{c}\,\partial_\mu c$$

$$- \frac{1}{\xi}(\partial_\nu A^\nu)\partial^2 c\,\delta\eta - \frac{1}{\xi}(\partial_\nu A^\nu)\delta\eta\,\partial^2 c$$

$$-\frac{1}{\xi}(\partial_\nu A^\nu)\left[\partial^2 c\,\delta\eta + \delta\eta\,\partial^2 c\right] = 0$$

since $\delta\eta$ and $c$ are Grassmannian

Now we expand $\bar{\Psi}'$ and $\Psi'$

$$\Rightarrow \quad \bar{\Psi}'(i\gamma - m)\Psi' = \left[\bar{\Psi}(x) + ie\bar{\Psi}(x)c(x)\delta\eta\right](i\gamma - m)\left[\Psi - iec\,\delta\eta\,\Psi\right]$$

$$= \bar{\Psi}(i\gamma - m)\Psi - ie\bar{\Psi}(i\gamma - m)c\,\delta\eta\,\Psi + ie\bar{\Psi}c\,\delta\eta(i\gamma - m)\Psi$$

$$= \bar{\Psi}(i\gamma - m)\Psi - ie\bar{\Psi}\left[i\gamma^\mu(\partial_\mu c)\delta\eta\,\Psi + i\gamma^\mu c\,\delta\eta(\partial_\mu\Psi) - mc\,\delta\eta\,\Psi\right]$$
$$+ ie\bar{\Psi}c\,\delta\eta(i\gamma - m)\Psi$$

$c$, $\delta\eta$, $\Psi$ are Grassmann variables $\Rightarrow c\,\delta\eta\,\Psi = -c\Psi\,\delta\eta = \Psi c\,\delta\eta$

$$= \bar{\Psi}(i\gamma - m)\Psi + e\bar{\Psi}\gamma^\mu\Psi\,\partial_\mu c\,\delta\eta - ie\bar{\Psi}c\,\delta\eta(i\gamma - m\Psi) + ie\bar{\Psi}c\,\delta\eta(i\gamma - m)$$

$$= \bar{\Psi}(i\gamma - m)\Psi + e\bar{\Psi}\gamma^\mu\Psi\,\partial_\mu c\,\delta\eta$$

$$\Rightarrow \bar{\Psi}'\gamma^\mu\Psi' = (\bar{\Psi} + ie\bar{\Psi}c\,\delta\eta)\gamma^\mu(\Psi - iec\,\delta\eta\,\Psi) = \bar{\Psi}\gamma^\mu\Psi + \underbrace{ie\bar{\Psi}c\,\delta\eta\,\gamma^\mu\Psi - ie\bar{\Psi}\gamma^\mu c\,\delta\eta\,\Psi}_{=0}$$

$$\Rightarrow \bar{\Psi}'\gamma^\mu\Psi' = \bar{\Psi}\gamma^\mu\Psi \quad \Rightarrow \quad \bar{\Psi}\gamma^\mu\Psi \text{ is invariant}$$

$$\Rightarrow \mathcal{L}' = \bar{\Psi}(i\gamma - m)\Psi + \underline{e\bar{\Psi}\gamma^\mu\Psi\,\partial_\mu c\,\delta\eta} - \frac{1}{4}F_{\mu\nu}F^{\mu\nu} - e\bar{\Psi}\gamma^\mu\Psi A_\mu - \underline{e\bar{\Psi}\gamma^\mu\Psi\,\partial_\mu c\,\delta\eta} + \frac{1}{2\xi}(\partial_\mu A^\mu)^2 + \partial^\mu\bar{c}\,\partial_\mu c$$

$$\mathcal{L}' = \bar{\Psi}(i\not{\partial} - m)\Psi - \frac{1}{4}F_{\mu\nu}F^{\mu\nu} - \frac{1}{2\xi}(\partial_\mu A^\mu)^2 + \partial^\mu\bar{c}\,\partial_\mu c = \mathcal{L}_{QED} - \frac{1}{2\xi}(\partial_\mu A^\mu)^2 + \partial^2\bar{c}\,\partial_\mu c$$

$\Rightarrow \mathcal{L}' = \mathcal{L} \quad \Rightarrow$ The Lagrangian $\mathcal{L}$ is invariant under the BRS transformations.